

Research Questions and Hypothesis

Question:

What buffer size achieves convergence?

Hypothesis:

Buffer size increases, performance converges at X

Experimental Variables

Independent Variable

Buffer sizes, two-stage grid search

Dependent Variables

15 metrics:
Time(3),
Statistical(2),
Frequency(10)

Controlled Variables

All other parameters fixed

CRISP-DM Phases

1. Data Understanding

- ERP-CORE N170
- 30 subjects × 30 channels
- 1024 Hz

2. Data Preparation

- Streaming emulation
- 1000ms windows
- High-pass filter (1 Hz)

3. Modeling

- 7 Buffer Sizes configs
- 5 baselines
- Fixed parameters

4. Evaluation

- 3-tier analysis
- Friedman + Nemenyi
- Convergence protocol

Three-tier Statistical Analysis

Tier 1: Baseline Comparison

12 methods (7 B_s + 5 baselines)
Is onEEGwaveLAD competitive?

Tier 2: Convergence Bound ID

Raw metric trajectories
Where do metrics stop changing?

Tier 3: Directional Significance

7 buffer configs only
Does B_s have significant effect?

Convergence Identification Protocol

Kneedle Elbow Detection

Median Aggregation

Absolute Criterion + Noise Floor